

Section 01



Industrial Security (Bachelor)

Section 01 – Introduction to Industrial Security

Introduction to Industrial Security

Lecture Notes

Department of Computer Science

- 1 Cyber-Physical Foundations
- 2 Context and History
- 3 Threat Setting and Misconceptions

By the end of this section you will be able to

- Define Operational Technology (OT) and contrast it with Information Technology (IT).
- Identify the core components of an Industrial Control System (ICS).
- Explain why the CIA triad is re-ordered for industrial environments.
- Place industrial security in its historical and economic context.
- Recognise common misconceptions about ICS security.

1

Cyber-Physical Foundations

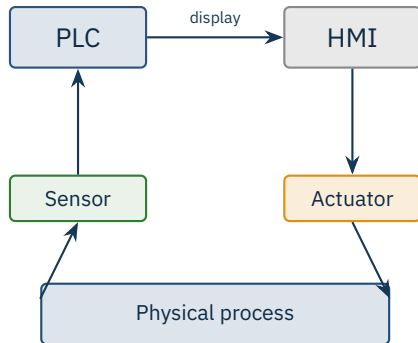
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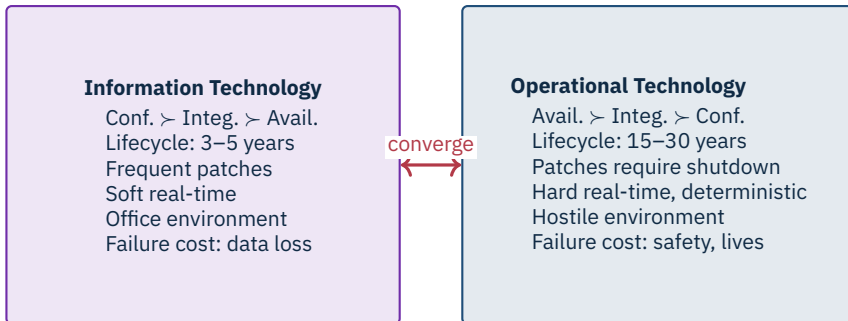
Definition

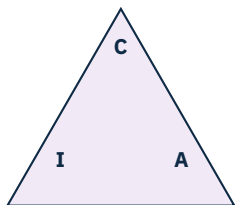
Industrial security is the protection of cyber-physical systems — computers that monitor and control machines in the physical world.

Sectors: power, water, oil & gas, manufacturing, transport, building automation.

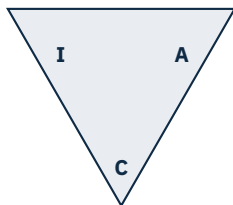
A breach has consequences in *atoms*, not just in *bits*.







IT priorities



OT priorities

Remember

In OT, **availability** and **safety** dominate the CIA triad. A blocked alarm can blind an operator to a runaway reactor.

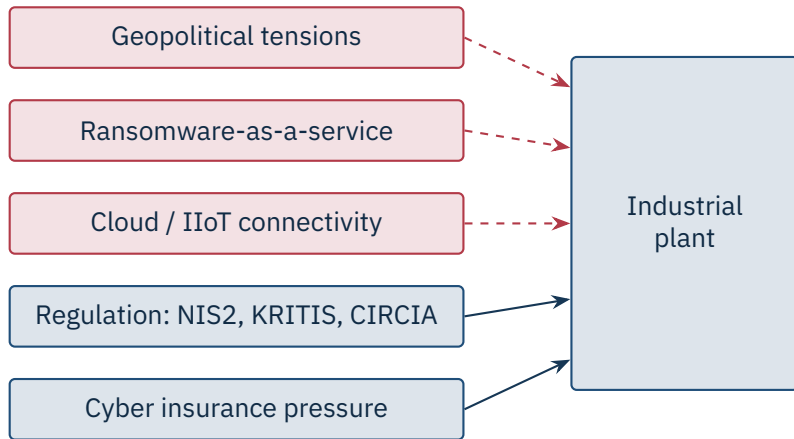
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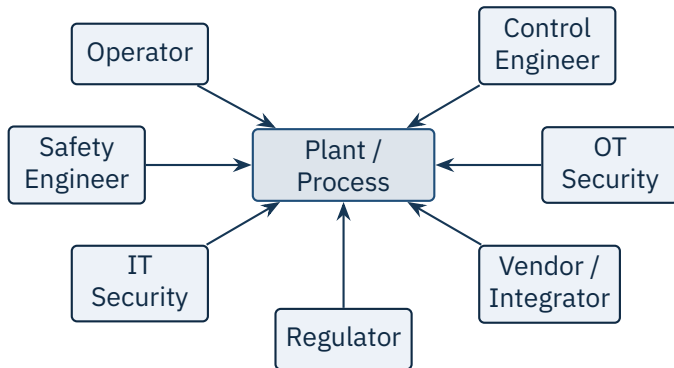
Context and History

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Each milestone made systems more capable – and more reachable.





Industrial security is a team sport across roles that rarely sat together historically.

3

Threat Setting and Misconceptions

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“We’re air-gapped, so we’re safe.”

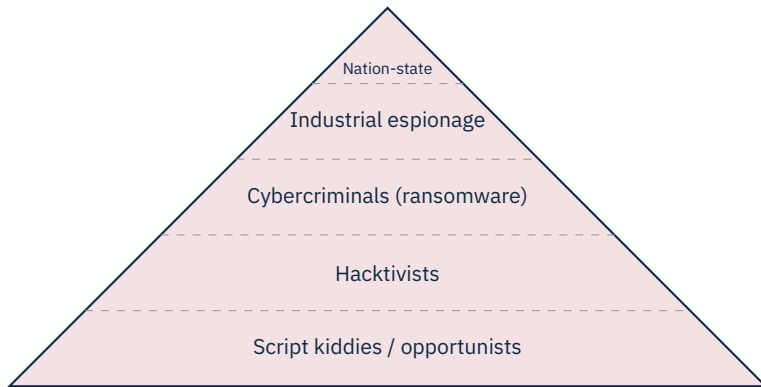
“Our protocols are obscure; nobody understands them.”

“We’ve never been attacked, so we’re fine.”

“IT tools will solve OT problems.”

“Patching solves everything.”

“Compliance equals security.”



Capability rises as we move up; volume of actors falls.

Colonial Pipeline \$4.4M ransom 5-day outage	Norsk Hydro \$71M loss 6-month recov.	Maersk (NotPetya) \$300M loss 10 days down	Stuxnet ~1000 centrifuges multi-year program
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Direct + indirect

Recovery costs typically exceed ransom by 5–10×. The lasting damage is reputation and regulatory.

Source: Colonial: FBI 2021; Norsk Hydro Q1 2019 report; Maersk Q2 2017 interim report; Stuxnet centrifuge count: D. Albright et al., ISIS, 2010.



New attack surface

Every cloud connector is a new ingress – and a new authentication challenge for legacy field devices.

- **OT Security Engineer** – design, deploy, monitor controls; bridges IT and OT
- **Control Engineer with security focus** – starts from process side
- **ICS Penetration Tester** – offensive; specialised employers
- **Compliance / Audit** – IEC 62443, NIS2 specialist
- **Incident Responder (OT)** – on-call, often consulting-side (Dragos, Mandiant)
- **Researcher** – vendor / academic; vulnerabilities, threat intel

Demand

Headhunter reports consistently rank OT security in the top three hardest-to-fill cyber roles globally.

ICS & SCADA fundamentals

Network architectures

Industrial protocols

Threat landscape

Vulnerabilities & assessment

Risk management

Defense in depth

Standards & governance

Incident response

★ Key Takeaway: Industrial security is different from IT security

- OT differs fundamentally from IT in goals, lifecycle, and tolerance for failure.
- Availability and safety dominate the CIA triad in industrial contexts.
- Adversaries today span from script kiddies to nation-state APTs.
- The next sections build the toolbox: components, networks, protocols, defenses.

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- Falliere, N., Murchu, L., Chien, E. *W32.Stuxnet Dossier*, v1.4. Symantec, Feb 2011.
- FBI. *Statement on the Colonial Pipeline attack*, May 2021.
- GAO. *Critical infrastructure protection: Federal authorities for the Colonial Pipeline response*, GAO-22-104746, 2022.
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- NIST. *Cybersecurity Framework 2.0*, Feb 2024.
- ENISA. *Threat Landscape*, annual.
- Knapp, E.D., Langill, J.T. *Industrial Network Security*, 2nd ed., Syngress, 2014.
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End of Section 01

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Questions and discussion